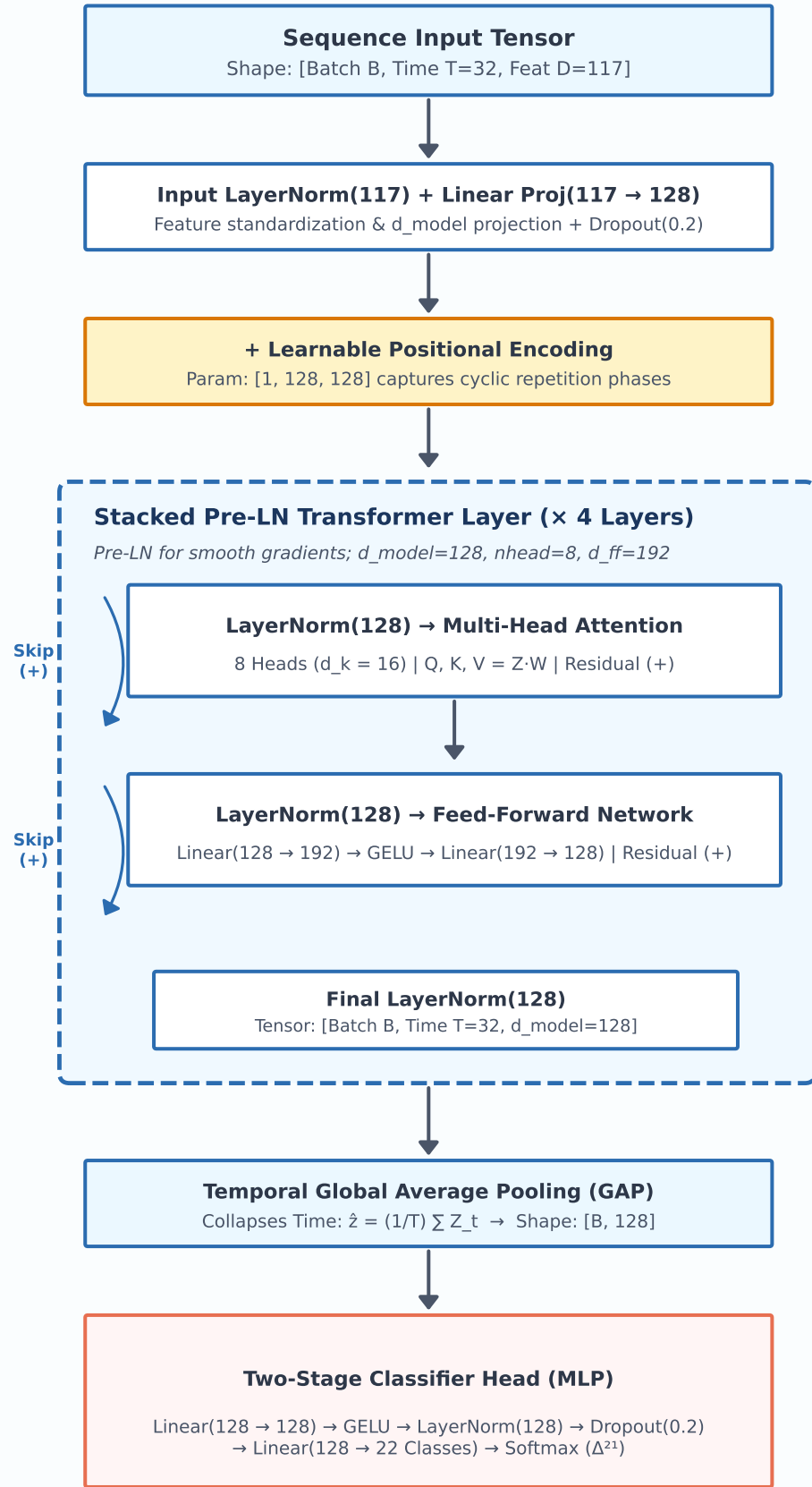


SkelGym Lightweight Single-Model Micro-Architectures (~350K-400K Parameters)

Calibrated Custom Backbones Built from Scratch for Privacy-Preserving Real-Time Gym Exercise Classification

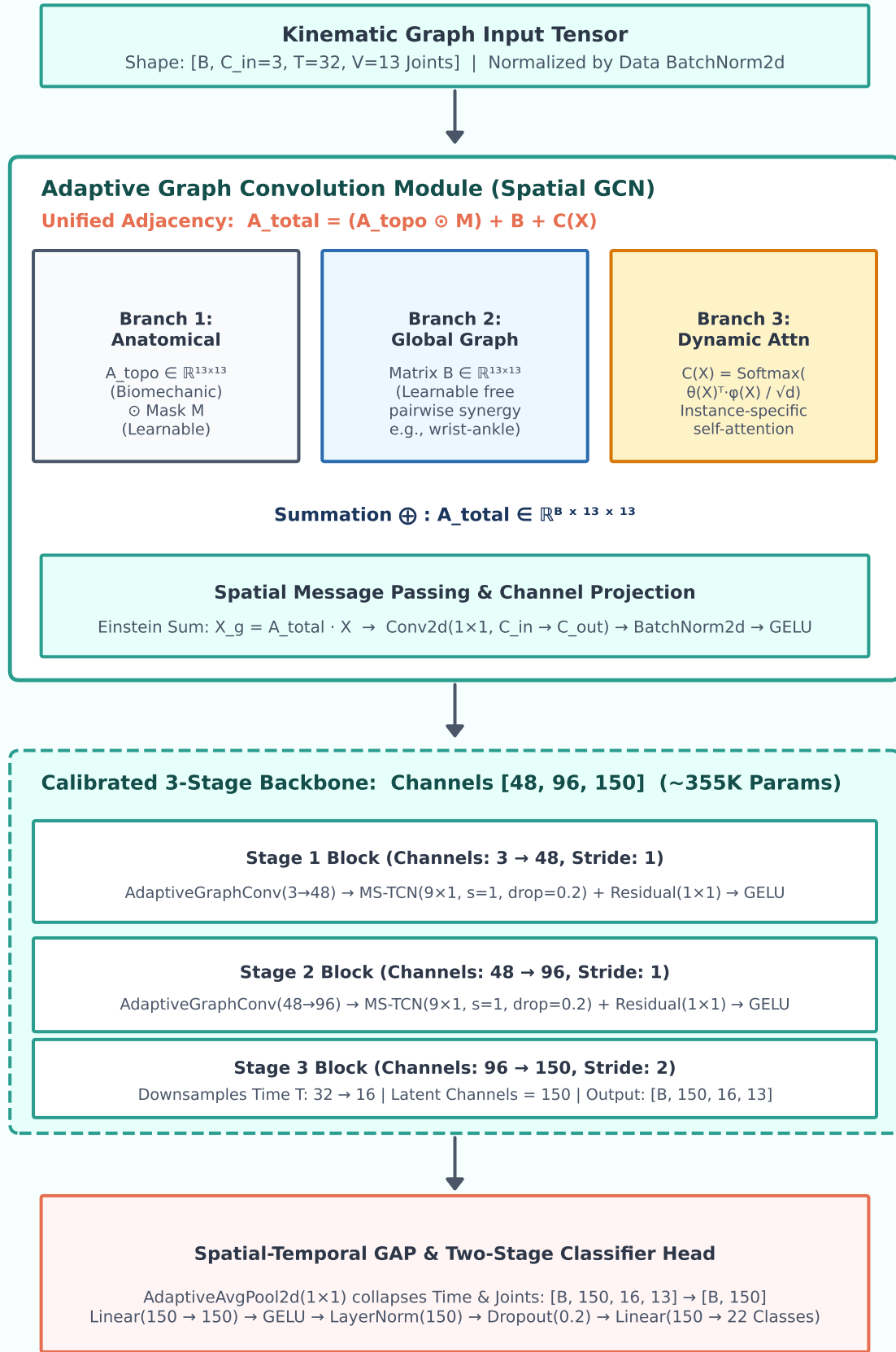
Panel A: Skeletal Transformer (Sequence Paradigm)

Input: Dense Sequence Mix (117-d) | 399K Params | 12.50 MFLOPs | Latency: 0.42 ms



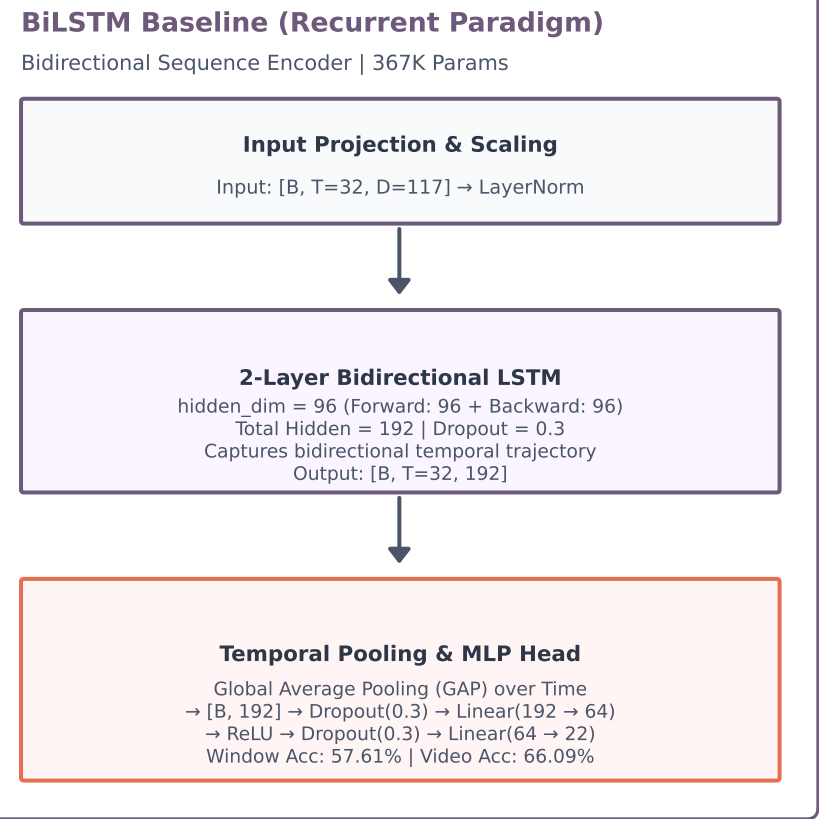
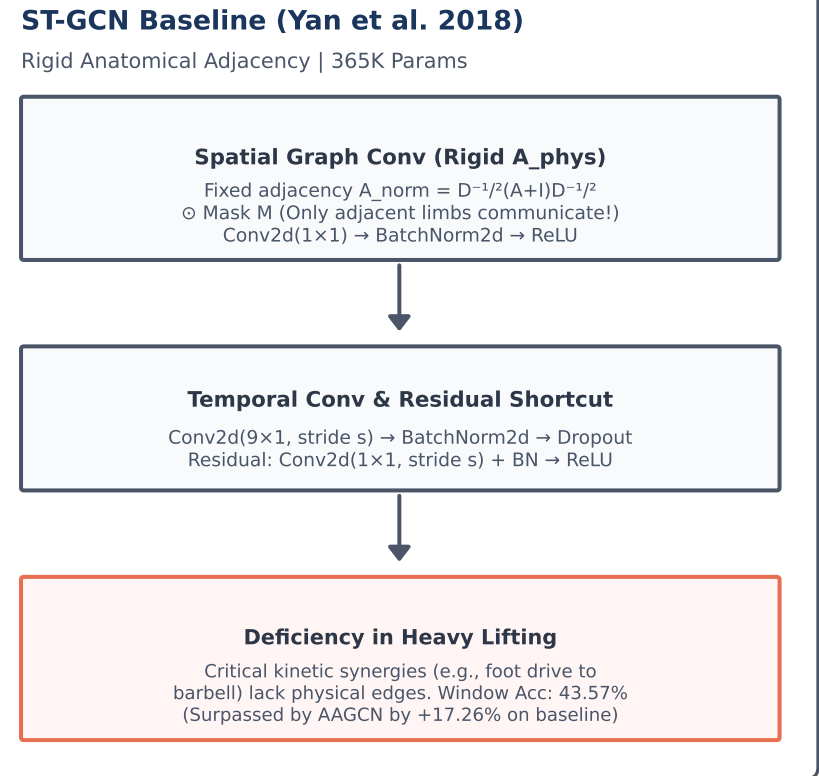
Panel B: Attention-Enhanced AAGCN Micro-Architecture

Input: Graph Stream (Joint/Bone/Motion) | 378K Params | 101.43 MFLOPs | Latency: 0.96 ms



Panel C: Comparative Baselines

Parameter-Calibrated (~350K) Reference Architectures



Note: All backbones are constrained to ~350K-400K parameters to ensure rigorous, fair comparative benchmarking on edge devices.